Ankle mobility

Mission: 3 clean, unassisted pistols on each leg. Today's limiter: a narrow, heels-down deep squat isn't there yet.

> **Today's win:** measure your knee-to-wall distance on each leg and text me the two numbers. Then do one round of knee-over-toe pulses. Five minutes.

Why ankles

At the bottom of a pistol your knee has to travel well past your toes with the heel down. If the ankle can't bend that far (dorsiflexion), you fall backward or fold forward. That's the big lean you've noticed. GMB names ankle dorsiflexion as the most common pistol limiter.[1]

1 · Measure: knee-to-wall test

1. Barefoot, face a wall in a half-kneeling lunge, front foot pointing straight at the wall.
2. Drive the front knee forward to touch the wall, **heel down**, knee over the middle toes.
3. Slide the foot back until you can only just touch. Measure big toe to wall in inches. Both legs.[2]
4. Retest weekly, same setup. A change of about ¾ in (1.9 cm) is real; smaller is measurement noise.[2]

Rough target: about 4 in (10 cm) or more. That's common field guidance for good dorsiflexion; under about 2½ in (6 cm) is restricted.[3] Some sources use 5 in.[2] Comparing your two legs matters more than any single number.

2 · Daily drills

Drill	Dose	How
Knee-over-toe pulses	10 slow reps + 20–30 s hold, each side, 1–2×/day	Half-kneeling, heel down, rock the knee forward past the toes and back; hold at the end range.[1] Optional: rest a light dumbbell on the knee for more stretch. If you feel a pinch at the front of the ankle rather than a stretch in the calf, the limit is likely in the joint[2]; a band looped low around the front of the ankle, pulling backward, is a common way to ease it.
Calf stretch, knee straight	30 s each side	Hands on the wall, back leg straight, toes forward, heel down; lean in. Targets gastrocnemius.[4]
Calf stretch, knee bent	30 s each side	Same, but bend the back knee, heel down. Targets soleus.[4]
Supported deep squat hold	1–2 min total/day, in chunks	Hold the tower upright, feet narrow, heels down, sink as deep as you can and breathe. GMB treats a deep two-leg squat as the pistol's prerequisite.[1]

Fits your short-session habit: do the pulses and squat hold on a tower visit, calf stretches whenever you stand up from the desk.

3 · Temporary heel lift

For pistol practice (Lesson 1 ladder), stand with your heel on a board or small plate, about 1 in high. Raising the heel reduces how far the ankle must bend and cuts forward trunk lean in squats.[5][6] It's a crutch, not the goal: as your knee-to-wall number climbs, step down to a thinner lift, then none. Log reps on a lift as "heel lift" in the variation.

Log it

Text Gopher your knee-to-wall inches each week (e.g. "KTW left 3, right 3.5"). Logged as type `knee-to-wall` in the progress log. Re-measured on the Oct 28 test day.

Check yourself

1. During the knee-to-wall test, your heel lifts as the knee touches. What then?

- count it as the score
- put your shoes on now
- slide the foot closer

2. What is a rough target for knee-to-wall distance?

- about 4 in or more
- about 1 in or more
- about 9 in or more

3. Which calf stretch targets the soleus?

- straight leg, heel up
- bent knee, heels down
- straight leg, toes up

4. Your knee-to-wall improves by an inch. What happens to the heel lift?

- raise it to 2 inches
- keep it for good now
- step to thinner lift

Read this

Primary source: Physiotutors: Weight Bearing Lunge Test (short, with video). Then the Royal Orthopaedic Hospital calf-stretch sheet.

Anything unclear? **Ask your teacher.** Send your first knee-to-wall numbers and I'll set your weekly retest.

Sources: [1] GMB Fitness, Pistol Squat Progression. [2] Physiotutors, Weight Bearing Lunge Test (citing Powden et al. 2015). [3] Movement Assessment Technologies, Weight-Bearing Lunge Test. [4] Royal Orthopaedic Hospital (NHS), Calf stretching. [5] Effect of Heel Lift Insoles on… Barbell Squats (PMC9312299). [6] Comparing the kinematics of back squats performed with different heel elevations, Human Movement 2021.